

End-to-End STEM Subject Classification

A leakage-safe, evidence-grounded machine-learning feasibility study

Final macro-F1: 0.8755
95% bootstrap CI: 0.8471 to 0.8996
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Portfolio report for applied ML, data science, analytics, and intelligent-manufacturing roles

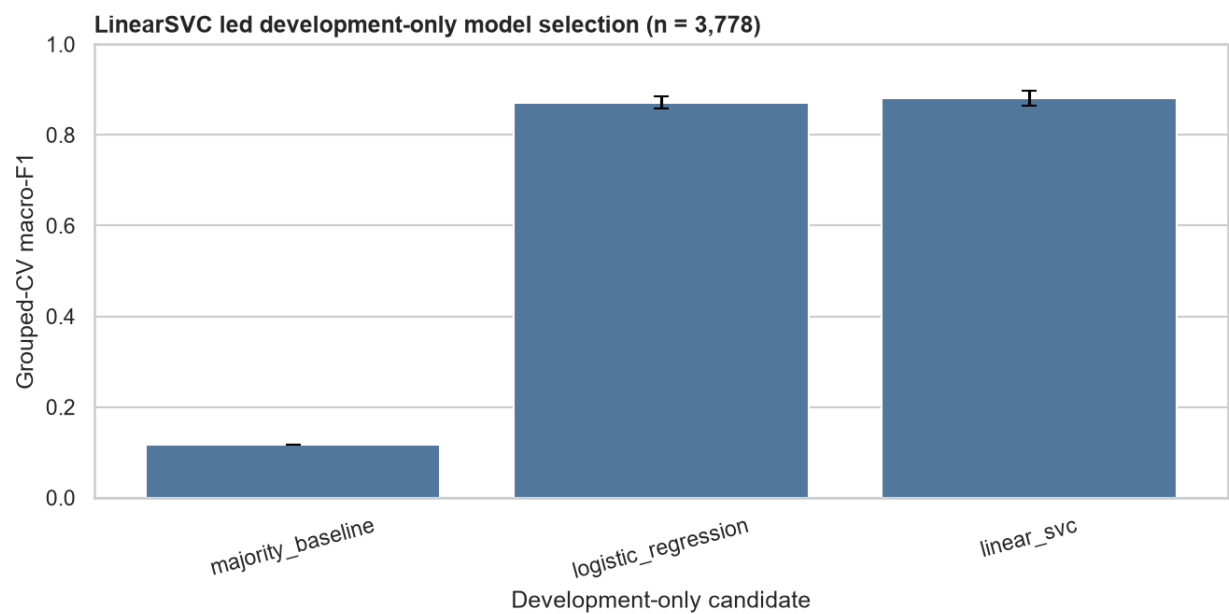
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1. Executive Summary

This study asks whether question text alone can route English STEM benchmark questions to mathematics, physics, chemistry, or biology. The locked linear_svc achieved macro-F1 **0.8755**, accuracy 0.8778, and a 95% stratified bootstrap interval of 0.8471 to 0.8996 on 630 isolated questions.

Decision: conditional go as a validated offline prototype. This supports continued validation, not production automation or claims about real students.



MMLU-Pro benchmark questions; final-test results are from the locked model only.

Figure 6. Development-only model comparison. Source: reports/tables/model_comparison.csv; verified.

2. Problem and Decision

The goal is a defensible ML workflow, not a frontend demonstration. A reviewer should be able to inspect data provenance, SQL, leakage controls, tuning, one-time test evaluation, errors, and limitations.

The feasibility criterion is substantial improvement over the majority baseline with stable grouped cross-validation and transparent failure analysis.

3. Data Source and Governance

Source: TIGER-Lab/MMLU-Pro at pinned revision b189ec765aa7ed75c8acfea42df31fdae71f97be. The source manifest records retrieval time, license note, row counts, and checksum.

Only normalized question text is used as a feature. Answers, answer options, explanations, and answer indices are prohibited. Raw data and model binaries are intentionally not committed.

4. Data Validation and EDA

Validation retained 4408 of 4499 rows and excluded 91 exact duplicates. The controlled data contains 4334 duplicate groups and 0 groups crossing the final split.

Class imbalance and source composition are measured before modeling. Source is highly associated with subject, creating shortcut and external-validity risk.

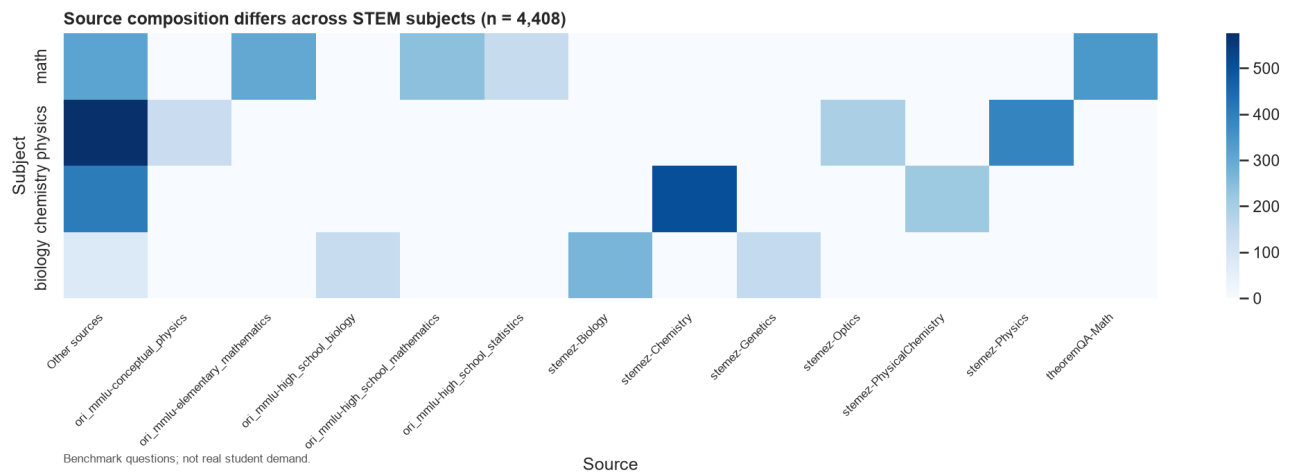


Figure 2. Subject by source composition. Source: SQL query 02; verified.

5. Leakage Control

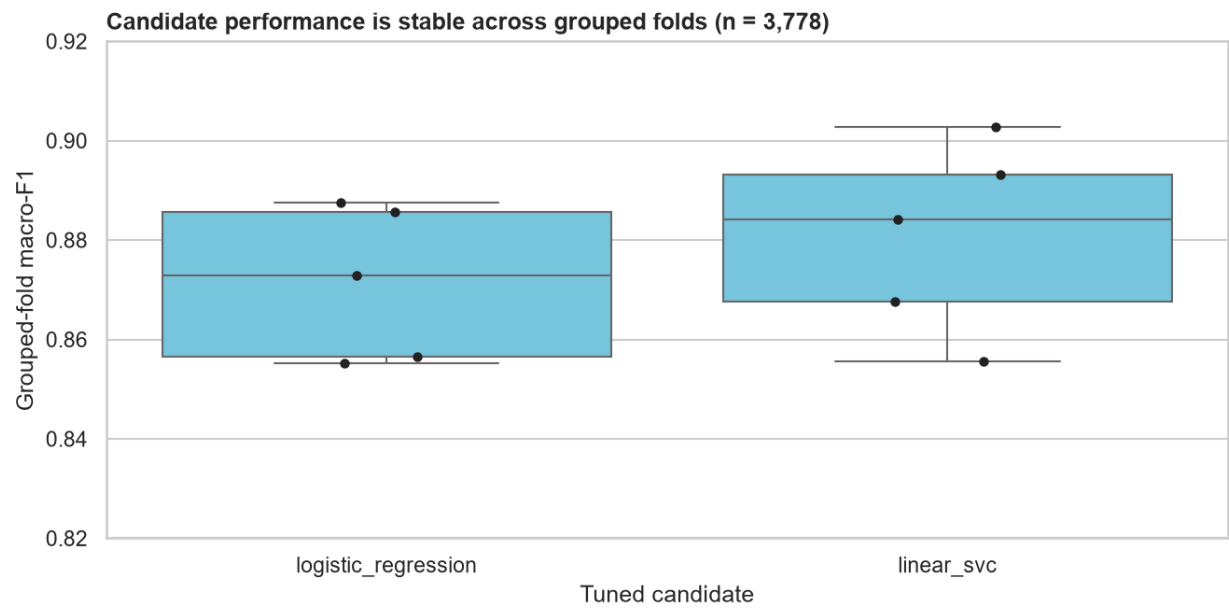
The final test fold was isolated before tuning. Exact duplicates were removed, near-duplicate groups were kept together, and TF-IDF fitting occurred inside each training fold.

Five-fold StratifiedGroupKFold on development data selected hyperparameters using macro-F1. The test set was opened only after the selection decision and hashes were persisted.

6. Modeling and Tuning

Candidates were majority baseline, TF-IDF plus Logistic Regression, and TF-IDF plus LinearSVC. LinearSVC was locked at CV macro-F1 0.8807 +/- 0.0171.

Locked parameters: {'classifier__C': 10.0, 'classifier__class_weight': None, 'tfidf__min_df': 1, 'tfidf__ngram_range': [1, 2]}. The final model designation was not changed after observing test results.



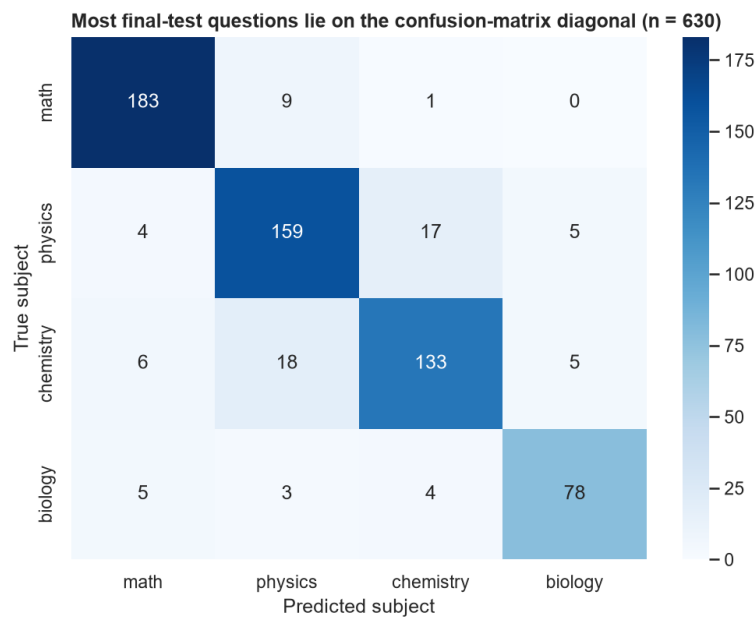
MMLU-Pro benchmark questions; final-test results are from the locked model only.

Figure 9. Grouped-fold variation. Source: reports/tables/cv_fold_variation.csv; verified.

7. Final Evaluation

Macro-F1 weights each class equally. Precision measures correctness among predictions; recall measures recovery among true items; weighted-F1 weights class F1 by support. The confusion matrix counts true/predicted pairs.

Final macro-F1 was 0.8755, weighted-F1 0.8774, and accuracy 0.8778. The uncertainty interval does not establish cross-domain performance.



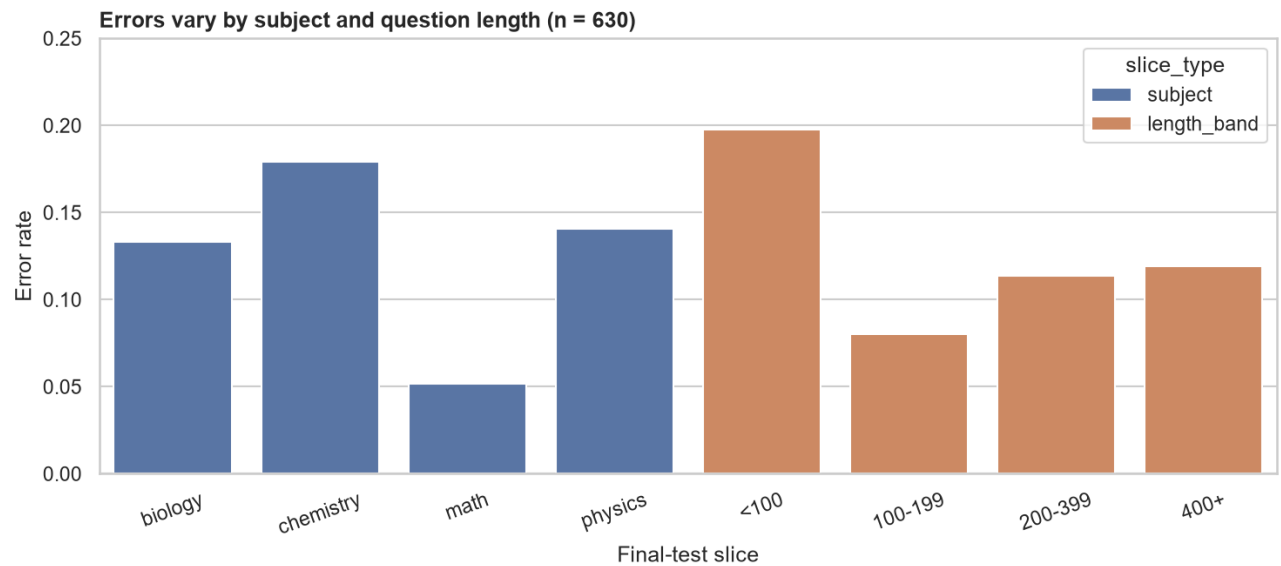
MMLU-Pro benchmark questions; final-test results are from the locked model only.

Figure 7. Locked final-test confusion matrix. Source: final_test_metrics.json; verified.

8. Error Analysis

Chemistry is weakest at F1 0.8391; mathematics is strongest at F1 0.9361. Short questions under 100 characters have the highest length-band error rate.

Representative cases and SQL slices make the errors auditable. Small per-source samples are descriptive and should not be overgeneralized.



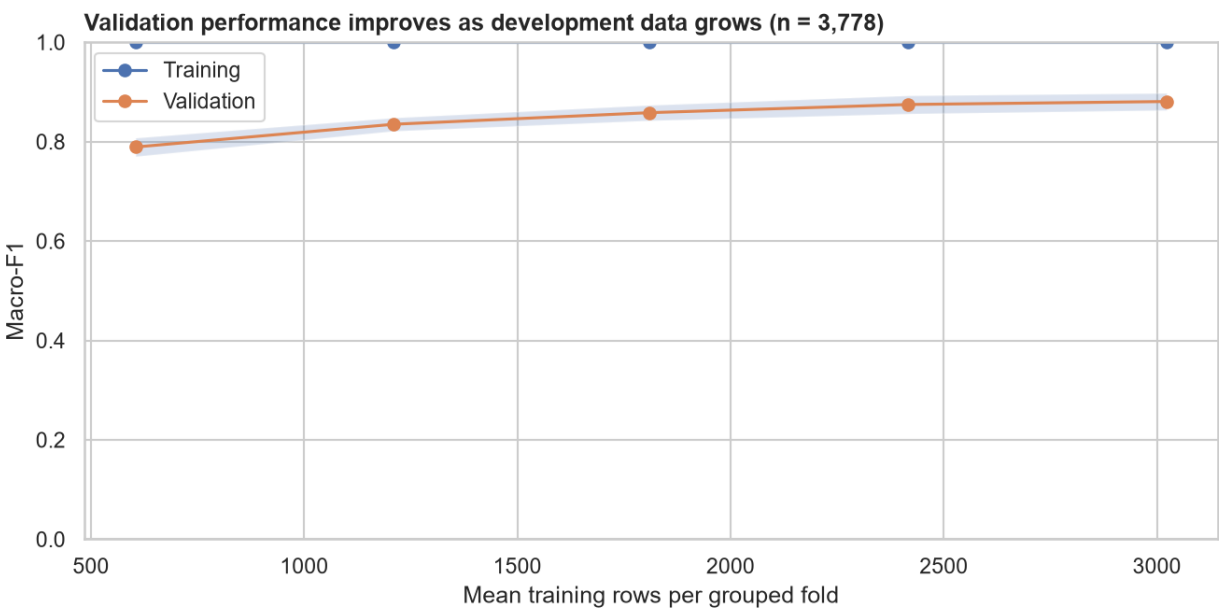
MMLU-Pro benchmark questions; final-test results are from the locked model only.

Figure 10. Error rates by subject and length band. Source: reports/tables/error_slices.csv; verified.

9. Features and Data Sufficiency

Linear coefficients reveal plausible domain vocabulary, but source-concentrated features flag shortcut risk. Feature importance is association, not causal explanation.

Grouped validation macro-F1 improves from 0.7891 at 20% development data to 0.8807 at full size, suggesting additional representative data may still help. No source contains at least 80 rows across all four subjects, so the predeclared source-holdout test is not feasible.



MMLU-Pro benchmark questions; final-test results are from the locked model only.

Figure 12. Grouped learning curve using development data only. Source: [reports/tables/learning_curve.csv](#); verified.

10. Model Capability Boundaries

Capability: validated offline prototype. It can demonstrate offline four-subject text routing under this benchmark split.

It cannot validate real student behavior, educational decisions, calibrated automation, production readiness, image or non-English inputs, open-set rejection, or guaranteed cross-source generalization. LinearSVC margins are not probabilities, so confidence/coverage automation is intentionally not claimed.

Confidence/coverage is intentionally not claimed (n = 630)

Not available

Locked LinearSVC margins are not calibrated probabilities.
A calibration protocol is required before threshold automation.

MMLU-Pro benchmark questions; final-test results are from the locked model only.

Figure 11. Confidence limitation. Source: reports/tables/confidence_coverage.csv; verified.

11. Reproducibility

Configuration, seed 42, pinned revision, manifests, controlled-data checksum, selection hash, SQL queries, figure checksums, tests, and commands are versioned.

Run `stem-analytics run-all --config configs/project.yaml` after installing the project. Raw data, SQLite, predictions, and model binaries rebuild locally and remain ignored by Git.

12. Intent Annotation Future Work

Intent classification is deferred because no trustworthy source label exists. The repository supplies a five-label taxonomy, a balanced 40-question pilot sampler, and validation rules only.

Future work requires independent double labeling, raw agreement, Cohen's kappa, adjudication, taxonomy revision when agreement is weak, and at least 20% double labeling before an 800-question study. No human annotation or intent model is claimed.

13. Appendix: Evidence and Metrics

Final-test per-class metrics

Subject	Precision	Recall	F1	Support
Math	0.9242	0.9482	0.9361	193
Physics	0.8413	0.8595	0.8503	185
Chemistry	0.8581	0.8210	0.8391	162
Biology	0.8864	0.8667	0.8764	90

Authoritative sources: artifacts/metrics/final_test_metrics.json, selection_decision.json, generated tables, manifests, SQL exports, and figure source register. All headline numerical consistency checks pass.